Submission pack

Words for the hackathon form and the video. Everything here matches `docs/AS-BUILT.md`; where the two differ, that page is right. Lines marked **(confirm)** are for me to check or fill before sending.

Project description (short)

The Vayu Project is a free app that turns public air-quality data into a daily decision for a Delhi family, school or society: is it a good time for a walk, is school safe, do we need a mask, are the windows better shut. It shows the air at the places you set (home, school, work, a park), what a person breathed over the day (Exposure), a world map of monitors, wind, fires and the colour between them, and an AIR Lens that turns a number into particles in front of the camera. Alerts for the next three days follow the GRAP stages, smog, stubble burning, rain and wind, and the Insights for a place say what that means for a school's assembly or a walk in a park.

Every number says whether it is **Live** (a named monitor, how far, how old) or an **Estimate** (a regional model). Nothing is advised from a reading older than six hours. A small measuring unit (ESP32-S3 and an SPS30 laser particle sensor) connects over Bluetooth and becomes a "Your room" card: the room's live air against the air outside, with the hours it measured counted in Exposure. A prototype air cleaner (a fan, a filter and a foam pre-filter in a box) is built and its outlet airflow measured at about 191 cfm; its CADR is not measured, it does not switch by itself.

It is all real data. The live app reads real monitors, weather and satellite fires. The recorded days (`/#/week`) is seven real days from the public archives played through the same app, labelled recorded. The only made-up days are in a separate prototype build that is never uploaded.

Live: https://d3lmq1mze2gup3.cloudfront.net

How it uses AWS

CloudFront and S3 serve the app; API Gateway and a Lambda serve `/v1`; DynamoDB caches and holds quotas; EventBridge Scheduler runs the harvest and the map's monitor job so visitors never call a provider; Bedrock (Amazon Nova Micro, with a guardrail) only rewords answers that fixed rules already decided; Polly reads answers aloud; Step Functions run the alert loop and "did it work?"; Secrets Manager, an SSM kill switch, throttles and daily caps keep it safe and cheap; everything is CDK in TypeScript. See `docs/architecture.md`, `docs/cost.md` and `docs/safety.md`.

Video script (3 minutes at most)

The shot list, the words to say, the console pages to cut to, the setup and what not to say are in `VIDEO.md`. It is the one source for the video.

The recorded days is built (`apps/web/src/week`, data made by `scripts/record-week.mjs`, see `docs/AS-BUILT.md`), and so is the animated architecture diagram: open `docs/architecture-animated.html` in a browser (five scenes; Space plays or pauses, arrow keys step, `#step=3` holds on a scene).

What each person did (confirm)

I built this alone (`CONTRIBUTIONS.md`). Every change goes through `make prove` (lint, types, tests, `cdk synth`, bundle check). I designed, wired, flashed and tested the measuring unit myself.

AI services in the app

- Amazon Bedrock, Nova Micro, with a guardrail: rewording the assistant's answers at run time. It has no tools and cannot act. See `docs/AGENT-BOUNDARIES.md`.
- Amazon Polly: reads answers aloud in English and Hindi. Hindi answers are fixed wording and never go through the model.

Feedback on AWS (confirm and edit)

Things we hit, in the order we hit them:

- CloudFront forwarding the viewer's `Host` header to API Gateway gives a 403; the origin request policy has to leave it out.
- The Bedrock guardrail's prompt-attack filter blocks instructions placed in the user turn. Putting the rewording instruction in the system prompt works.
- `cdk deploy` behind a corporate network needs the system certificate store (`NODE_OPTIONS=--use-system-ca`).
- Scheduled jobs that call a provider need their own time budget inside the function's time limit, and a schedule that never runs needs its own alarm; neither is a default.
- Lambda egress addresses differ from a laptop's, which is why the wind and air model are asked from the API once for everyone rather than from each browser. Open-Meteo counts each point of a many-point request as a call against the address's daily allowance, so a map's wind grid used it up quickly.
- Bedrock needs the regional inference profile id (`apac.amazon.nova-micro-v1:0`), not the bare model id, in ap-south-1.

Honest limits

- The measuring unit reads a real sensor and reaches the app, but its accuracy has **not** been compared with a monitor, so it is a low-cost sensor shown beside the nearest monitor, never "scientific-grade". See `docs/claims.md`.
- The cleaner's 191 cfm is airflow at the outlet, not CADR: the smoke-decay test has not been run, and it does not switch by itself. Rules I13 and I16 are untested on a device.
- Telegram alerts are built and tested against fakes; live scheduled sends are off and there is no in-app screen to link a chat.
- Bluetooth: Chrome on a computer is tested; Android Chrome is not; iPhone Safari cannot.
- The ₹2,00,00,000 (a station) and ₹4,000 (the unit) figures have no source in the repository yet: do not say them without one.
- The GRAP stages in the recorded days are our estimate for those days, not a record of the official order. The estimate is the last 24 hours' average AQI on India's scale, checked against the monitors' own PM10 (no day in the wrong stage on 24 station-days); an hour can be worse than the day's average and the app says so.
- OpenAQ rejected our key (401) late on 10 October and a new one was put in on 11 October; the stations job then read 7,268 monitors. If it is rejected again, the map's colour comes from the regional air model and says so.